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Research article

The impact of cyber threats on environmental, social, and governance performance

Kaouther Chebbi

Finance Department, School of Business, King Faisal University, Al-Ahsa, Saudi Arabia

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ABSTRACT

This study investigates the relationship between cybersecurity risk and corporate Environmental, Social, and Governance (ESG) performance, with a particular focus on the moderating role of CEO duality in shaping strategic responses to digital threats. Drawing on a large sample of 14,036 firm-year observations from U.S.-listed firms between 2005 and 2018, we find that heightened exposure to cybersecurity risk is significantly associated with improved ESG performance, particularly in the environmental dimension. Our results further indicate that this positive association is amplified in firms where the CEO also serves as the board chair, suggesting that centralized leadership structures enhance strategic agility and resource allocation in response to emerging risks. We address potential endogeneity concerns using robust econometric techniques, including two-stage least squares (2SLS) and Heckman selection models, confirming the reliability of our findings. These insights extend current ESG frameworks by incorporating digital resilience as an emerging driver of sustainability performance and enrich governance theory by highlighting the contextual influence of CEO duality on risk management strategies.

The study offers practical implications for corporate leaders, investors, and policymakers. Firms can leverage cybersecurity risk management as a catalyst for strengthening ESG practices, particularly in sectors exposed to high digital risks. Governance structures that enable swift decision-making may enhance firms' capacity to align cybersecurity and sustainability goals. Policymakers are encouraged to consider integrating cybersecurity metrics into ESG disclosure frameworks to promote more comprehensive corporate accountability and resilience in the digital era.

1. Introduction

The rapid digitization of business operations has elevated cybersecurity from a technical concern to a central corporate governance issue (Amoo et al., 2024; Yusif et al., 2024; Cortez and Dekker, 2022). The increasing frequency and severity of cyberattacks have transformed cybersecurity risk into a strategic challenge, with global cybercrime losses projected to reach $10.5 trillion annually by 2025 (Uddin et al., 2020a,b; Umoga et al., 2024). These threats jeopardize financial stability and affect stakeholder perceptions regarding corporate transparency, accountability, and resilience (Gold, 2023; Handayani, 2019). In this context, Environmental, Social, and Governance (ESG) performance has emerged as a critical metric for assessing corporate responsibility and sustainability, shaping how firms are evaluated by investors, regulators, and the public (Khamisu et al., 2024; Kuzey et al., 2023).

Recent studies highlight the growing interdependence between cybersecurity risk management and ESG performance, suggesting that organizations increasingly view robust cybersecurity practices as integral to demonstrating good governance and long-term sustainability (Schaffner et al., 2023; Morales-Sáenz et al., 2024; Battaglia et al., 2025). From a theoretical perspective, stakeholder theory emphasizes that firms must respond to the expectations of diverse constituencies—particularly in times of heightened digital risk (Mu et al., 2024; Gold, 2023). Similarly, signaling theory suggests that firms may use ESG disclosures as a mechanism to communicate their digital resilience and operational integrity to external stakeholders (Biswas et al., 2025). Despite these insights, the empirical link between cybersecurity risks and comprehensive ESG performance remains underexplored, especially regarding its multidimensional impact across environmental, social, and governance domains.

This study contributes to the existing literature by offering one of the first empirical examinations of how cybersecurity risk influences aggregate ESG performance and subcomponents. Drawing on 14,036

E-mail address: kchebbi@kfu.edu.sa.

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firm-year observations from S&P 1500 companies between 2005 and 2018, our analysis reveals a significant positive relationship between cybersecurity risk exposure and ESG performance, indicating that firms tend to strengthen their ESG practices in response to increased digital threats. Moreover, we introduce CEO duality as a moderating variable, examining how leadership structure affects the relationship between cybersecurity risk and ESG outcomes. This adds a new dimension to the ongoing debate on the implications of CEO duality for corporate governance, particularly in the context of green governance and digital risk management (Chebbi et al., 2024; Abdullah et al., 2024; Panda and Leepsa, 2017).

Our findings further demonstrate that firms facing higher cybersecurity risks exhibit improved performance across all three pillars of ESG: environmental, social, and governance dimensions. Additionally, CEO duality significantly moderates the relationship between cybersecurity risk and ESG performance. Specifically, in firms where the CEO also serves as board chair, the positive effect of cybersecurity risk on ESG performance is amplified. These results suggest that centralized leadership structures can enhance responsiveness to digital threats through more coordinated and strategic ESG initiatives. We confirm the robustness of our findings using alternative measures of cybersecurity risk, as well as econometric techniques such as two-stage least squares (2SLS) and Heckman selection models to address endogeneity and self-selection bias.

This study makes several significant contributions to the growing literature on cybersecurity risk management, corporate governance, and ESG performance. First, it extends the current understanding of how cybersecurity risks—increasingly recognized as a critical strategic challenge—influence firm-level Environmental, Social, and Governance (ESG) performance. While prior research has examined the effects of data breaches on corporate social responsibility, this is one of the first empirical studies to investigate the broader relationship between cybersecurity risk and comprehensive ESG performance, including its environmental, social, and governance subcomponents [Amoo et al., 2024; Yusif et al., 2024; Cortez and Dekker, 2022]. Second, the study introduces CEO duality as a novel moderating factor in the cybersecurity-ESG nexus, offering fresh insights into how corporate leadership structures shape organizational responses to digital threats (Chebbi et al., 2024; Abdullah et al., 2024). Third, our findings offer practical implications for corporate leaders, investors, and policymakers, demonstrating how ESG strategies can serve as both a reputational safeguard and a strategic response to evolving cybersecurity challenges (Handayani, 2019; Gold, 2023). In sum, this research enriches theoretical and empirical discussions at the intersection of digital risk, sustainability, and corporate governance, offering a timely contribution to the ongoing discourse on responsible business practices in the digital age [Schaffner et al., 2023; Morales-Sáenz et al., 2024; Battaglia et al., 2025].

The remainder of the paper is structured as follows: Section 2 presents the conceptual framework and develops the hypotheses. Section 3 outlines the data, variables, and econometric methodology. Section 5 presents empirical results, robustness checks, and sectoral insights. Section 6 concludes with a discussion of the theoretical and practical implications, limitations, and avenues for future research.

2. Conceptual framework and hypothesis development

2.1. Cybersecurity and ESG integration

The digital transformation of business operations has elevated cybersecurity to a core concern in corporate governance. The increasing frequency and intensity of cyberattacks—expected to cause global losses of $10.5 trillion annually by 2025 (Uddin et al., 2020a,b)—have prompted firms to integrate cybersecurity risk management into their broader ESG strategies. This integration reflects growing stakeholder expectations around resilience, transparency, and sustainable corporate behavior (Schaffner et al., 2023; Morales-Sáenz et al., 2024; Battaglia et al., 2025; Stef et al., 2025).

From the stakeholder theory perspective, firms must address the expectations of multiple constituencies—including investors, customers, employees, and regulators—by demonstrating accountability and proactive risk mitigation (Handayani, 2019; Gold, 2023; Mu et al., 2024). Cybersecurity threats heighten stakeholder scrutiny (Bansal and Axelton, 2024), prompting organizations to enhance their ESG disclosures as a strategic and reputational safeguard (Lee et al., 2022; MA et al., 2024). Enhanced ESG performance communicates responsible governance, supporting trust and long-term value creation (Fafaliou et al., 2022; Zhang et al., 2025).

Additionally, signaling theory underscores how firms convey information about their internal quality and risk management capabilities through external indicators such as ESG disclosures (Biswas et al., 2025). In the context of cybersecurity threats, firms can utilize enhanced ESG reporting to signal their digital resilience and operational integrity to investors and the market (Zhou et al., 2022). This signaling helps differentiate firms in competitive markets where intangible assets, such as trust and transparency, increasingly influence valuations.

Empirical studies confirm that organizations often enhance environmental and social efforts following data breaches as a means of reputational recovery (Chamberlain et al., 2025). Environmental goals, such as protecting critical infrastructure or promoting energy efficiency, rely on secure digital systems. Likewise, social outcomes—such as data privacy and stakeholder protection—are strengthened by effective cybersecurity practices. From a governance standpoint, cybersecurity oversight is now a proxy for board competence, and firms that embed cyber resilience into their ESG reporting tend to be viewed as more transparent and accountable (Fitzgerald, 2018; Nolan et al., 2019; Héroux and Fortin, 2024).

In sum, theoretical and empirical evidence suggest that cybersecurity risk functions as a strategic driver of ESG performance, particularly when firms respond with transparent, credible, and stakeholder-focused initiatives. In light of the foregoing discussion, we propose the following first hypothesis.

H1. Cybersecurity risk increases ESG performance

2.2. The role of CEO duality

CEO duality, where the CEO also serves as the board chair, represents a key governance configuration that may influence how firms manage cybersecurity risks and their broader environmental, social, and governance (ESG) commitments. This structure can both concentrate power and streamline decision-making, resulting in mixed governance outcomes that depend on the context.

Agency theory argues that CEO duality reduces agency costs by aligning the interests of management and the board, facilitating faster and more unified decision-making (Chebbi et al., 2024; Abdullah et al., 2024). In the face of cybersecurity threats, such cohesion allows firms to respond quickly with well-resourced ESG initiatives. The centralized authority of a dual-role CEO may enhance the prioritization of digital resilience and sustainability across the organization (Panda and Leepsa, 2017).

Conversely, critics suggest that CEO duality may undermine board independence and accountability, potentially leading to governance lapses (Oyewo, 2023). This concern is particularly relevant in contexts where digital risks require multi-stakeholder engagement and oversight.

Stewardship theory offers a more optimistic view. It posits that executives with dual roles act as stewards of the firm's long-term interests, integrating ethical governance with strategic foresight. In cybersecurity contexts, a CEO-chairperson may champion ESG-linked initiatives not merely for compliance but as part of a mission to enhance the firm's reputation, resilience, and stakeholder relationships (Ali et al., 2022; Mahran and Elamer, 2024; Shabbir et al., 2024). Dual-role leaders may

view cybersecurity governance not just as a risk management tool, but as a cornerstone of corporate sustainability and ethical leadership.

Overall, both theoretical perspectives suggest that CEO duality can play a crucial role, either enabling or constraining, in how cybersecurity threats influence ESG strategies. CEO duality may amplify the positive impact of cybersecurity awareness by aligning leadership structures with strategic, sustainability-oriented action. We propose the following second hypothesis.

H2. CEO duality moderates the nexus between Cybersecurity risk and ESG performance

3. Research methods

3.1. Data and sample

Our study leverages a comprehensive dataset of firms within the S&P 1500 Index. To address concerns regarding potential biases in ESG scores, such as those highlighted by Drempetic et al., (2020), which suggest that larger firms may be better positioned to communicate their ESG activities, thereby inflating their scores, we focus exclusively on large firms within the index that have established ESG ratings from Bloomberg. The cybersecurity risk measure used in our analysis is based on the text-based Cyber score developed by Lattanzio and Ma (2023). We obtained this data externally and merged it with our datasets. Financial data are sourced from Compustat, and all datasets are combined and cleaned of missing observations, resulting in a final sample of 14,036 firm-year observations. We selected the period from 2005 to 2018 to align with the availability of consistent ESG data from Bloomberg while ensuring an overlap with the full range of cybersecurity risk scores. This extended time frame enhances the reliability and generalizability of our findings.

3.2. Variables

3.2.1. Predicted variable

The cybersecurity risk score reflects an organization's exposure to cybercrime and the associated liabilities stemming from IT risks, acting much like a credit score for cybersecurity by offering insights into the effectiveness of an organization's internal and external IT asset management practices. We employ the cybersecurity score developed by Lattanzio and Ma (2023), a text-based measure that assesses firms' exposure to cybersecurity risk by systematically scanning 10-K filings from 1997 to 2018 for disclosures containing cybersecurity-related keywords. The index utilizes a comprehensive glossary derived from the National Initiative for Cybersecurity Careers and Studies as well as terms used in prior research, including Florackis et al. (2023). It also incorporates a range of fixed effects, such as state-year, industry-year, and firm-level controls, making it a robust and reliable proxy for firms' expected, or *ex-ante*, cybersecurity risk. We favor this measure over alternatives proposed by Florackis et al. (2023) and Jiang et al. (2024), which are limited to data starting in 2007 due to their reliance on reported cyberattacks and mandated cybersecurity disclosures. These shorter timeframes do not align well with our use of the Bloomberg ESG database, which begins in 2005. Lattanzio and Ma's (2023) broader temporal coverage enables a more comprehensive analysis of the relationship between cybersecurity risk and financial outcomes, resulting in more generalizable and representative findings.

3.2.2. Predictor variable

We utilize Bloomberg ESG disclosure pillar scores, building upon prior research (Sassen et al., 2016; Cornell, 2021; Reber et al., 2022; Bax et al., 2023; Tsang et al., 2023). An ESG score is derived from a company's reported data on its environmental impact, social responsibility, and governance practices. The ESG score equally weights all pillars, ranging from 0.1 (minimum disclosure) to 100 (comprehensive disclosure encompassing all data points captured by Bloomberg). The ESG score gauges a company's environmental, social, and governance transparency. It is a composite score made up of three sub-scores. The Environmental Score (ENV) focuses on a company's carbon footprint and other environmental impacts, while the Social Score (SOC) reflects a company's relationship with its employees and the broader society. The Governance Score (GOV) assesses the company's board structure and its commitment to shareholder rights.

3.2.3. Moderator variable

CEO duality is a dummy variable that takes the value of one if the CEO is the board chair and zero otherwise.

3.2.4. Control variables

This study employs a set of control variables identified in the extant literature as relevant determinants of ESG performance (Sabbaghi, 2022; Wang et al., 2023; Chebbi, 2024; Zeng et al., 2024). We include the size of the firm, which is calculated as the natural logarithm of total assets and return on assets; leverage, which is calculated as the ratio of total debt to total assets; tangible assets, which is calculated as the ratio of tangible assets to total assets, cash which is calculated as the ratio of cash flows to total assets, market capitalization which is calculated as the number of shares multiplied by the stock price, R&D which is calculated as the ratio of research and development expenses to total assets and growth which is the growth rate of sales.

3.3. Models

Equation (1) examines the association between cybersecurity risk and ESG performance, accounting for control variables

$$ESG = \alpha 0 + \alpha 1\ CYBERISK + \alpha 2\ SIZE + \alpha 3\ ROA + \alpha 4\ LEV + \alpha 5\ TANGIBLE + \alpha 6\ CASH + \alpha 7\ MARKETCAP + \alpha 8\ R\&D + \alpha 9\ GROWTH + YEARDUMMIES + INDUSTRYDUMMIES + STATEDUMMIES + \mu i,t \quad (1)$$

Equation (2) examines the moderating role of CEO duality on the association between cybersecurity risk and ESG performance, accounting for control variables

$$ESG = \alpha 0 + \alpha 1\ CYBERISK + \alpha 2\ CEODUALITY + \alpha 3\ CYBERISK * CEODUALITY + \alpha 4\ SIZE + \alpha 5\ ROA + \alpha 6\ LEV + \alpha 7\ TANGIBLE + \alpha 8\ CASH + \alpha 9\ MARKETCAP + \alpha 10\ R\&D + \alpha 11\ GROWTH + YEARDUMMIES + INDUSTRYDUMMIES + STATEDUMMIES + \mu i,t \quad (2)$$

This study employs fixed effects regression models to estimate the causal impact of cybersecurity risk on ESG strategies, as well as the moderating influence of CEO duality. This methodological approach is particularly appropriate when unobserved, time-invariant firm-specific characteristics may confound the relationship between cybersecurity risk and ESG performance. By incorporating firm and year fixed effects, our model controls for both observed and unobserved heterogeneity that could otherwise bias the estimates. This allows us to isolate the effect of CYBERRISK on ESG outcomes while holding constant stable, firm-specific attributes such as corporate culture, governance structure, or long-term strategic orientation. The use of fixed effects enhances the robustness of our inferences by focusing on within-firm variation over time, thereby providing more reliable estimates of the dynamic relationship between cybersecurity exposure and sustainability-oriented corporate strategies.

4. Empirical results and discussion

4.1. Univariate analysis

Descriptive statistics provide a snapshot of the data in Table 1, offering insights into the central tendency and variability of each variable. ESG scores, encompassing environmental, social, and governance

Table 1
Descriptive statistics.

Variables	N	Mean	STD	25th percentile	Median	75th percentile
ESG	14,036	42.1278	12.8991	32.9187	38.5289	49.6859
ENV	14,036	41.6617	17.6581	20.8759	34.3929	43.1034
SOC	14,036	42.5284	14.7001	22.7719	35.8276	46.5625
GOV	14,036	54.2893	7.4951	51.7857	51.7857	55.3571
CYBERISK	14,036	0.0668	0.0477	0.0000	0.0776	0.1267
SIZE	14,036	22.1523	1.6701	20.9856	22.0316	23.2081
ROA	14,036	0.0821	0.0977	0.0317	0.0729	0.1241
LEV	14,036	24.0384	20.3849	7.8007	21.8222	35.6023
TANGIBLE	14,036	0.7948	0.2121	0.6521	0.8666	0.9811
Cash	14,036	0.0536	0.0827	0.0128	0.0500	0.0935
MARKCAP	14,036	1.3431	1.3761	0.5250	0.9615	1.6842
R&D	14,036	0.0218	0.0463	0.0000	0.0000	0.0220
Growth	14,036	9.8318	33.1062	−0.0219	6.4246	14.7153

aspects, are rated on a scale ranging from 0 to 100. The closeness between the means and medians of the ESG data indicates a normal distribution with minimal influence from extreme values, strengthening the reliability of our regression results. These statistics are close to those of Aabo and Giorici (2023). The mean CYBERISK is almost 0.07, with a standard deviation of 0.0477. These statistics are close to those of Lattanzio and Yue (2023) Lattanzio and Ma (2023)Return on Assets (ROA) averages at 0.082 with a moderate standard deviation, while Leverage (LEV) and Market Capitalization (MARKCAP) demonstrate broader spreads around means of approximately 24 and 1.34, respectively. Tangible assets (TANGIBLE) have an average ratio of 0.795 with moderate variability. Cash ratios are generally low, averaging 0.054 with moderate variability. Spending on Research and Development (R&D) is generally low, with most companies reporting minimal expenditure. Growth rates (Growth) exhibit a high average of 9.832 %, with considerable variability ranging from negative values to high positive percentages.

Table 2 displays the correlation coefficients between all variables used in our baseline regressions (Equation (1)). The table indicates that ESG has a strong positive correlation with ENV, SOC, and GOV. This indicates that these variables tend to move in the same direction, meaning higher ESG scores are associated with higher environmental, social, and governance scores. ESG exhibits a positive correlation with CYBERISK. This suggests that cybersecurity risk scores (CYBERISK) tend to increase, and companies improve their ESG performance (higher ESG scores). We incorporate a comprehensive set of control variables in our regression analysis to mitigate the influence of spurious correlations. We check for multicollinearity among the independent variables before conducting the regression analysis. Table 2 shows the correlation coefficients between all variables. None of the correlations exceed the threshold of 0.50, as suggested by Hair et al. (2019). We calculated variance inflation factors (VIFs) below the concerning threshold of 10 to further assess multicollinearity. This indicates that multicollinearity is not a concern in our analysis.

Fig. 1 illustrates a significant finding of our research: a positive correlation between cybersecurity risk and ESG score. The data suggests that firms with higher cybersecurity risk tend to invest more in ESG activities, reinforcing the importance of cybersecurity in driving sustainable business practices.

4.2. Multivariate analysis

4.2.1. Cybersecurity risk and ESG performance

Table 3 presents the intriguing results of our analysis, estimated using Equation (1) with CYBERRISK as the key independent variable. The findings unveil a statistically significant positive association between cybersecurity risk and a firm's ESG performance. This can be gleaned from the estimated coefficient of CYBERISK, which is a substantial 4.107. Furthermore, the accompanying t-statistic of 3.31 indicates significance at the 1 % level, solidifying the robustness of this relationship. These results bolster our initial prediction, demonstrating that companies facing higher cybersecurity risk levels tend to exhibit demonstrably better ESG performance.

The analysis underlines the positive relationship between cybersecurity risks and ESG performance. It demonstrates that firms exposed to higher cybersecurity threats often implement enhanced governance practices to improve their resilience and align with sustainability goals. Research indicates that cybersecurity risks pressure firms to strengthen their governance structures, enhancing accountability and overall performance (Shackelford et al., 2023; Mortimer, 2021). Companies that adopt proactive cybersecurity measures often experience significant improvements in governance quality, which correlates with better ESG performance. This suggests that cybersecurity risks serve as catalysts for comprehensive governance strategies, reinforcing the integration of risk mitigation practices into ESG frameworks. Furthermore, the financial and operational damage from cyber incidents compels firms to adopt robust measures that align with their long-term sustainability objectives. The findings suggest that cybersecurity is not merely a standalone issue but a critical governance indicator, necessitating a greater corporate focus on resilience and strategic alignment (Shackelford et al., 2023; Mortimer, 2021).

Another potential explanation lies in the heightened awareness of the reputational risk associated with cyberattacks (Gwebu et al., 2018; Confente et al., 2019; Makridis, 2021). Companies exposed to significant cybersecurity threats may be more proactive in implementing robust environmental, social, and governance (ESG) practices. This could serve as a defensive measure, mitigating the potential damage to their reputation in the event of a successful cyberattack. Various studies support this notion, suggesting that firms with strong social and environmental practices experience less reputational harm (Jeffrey et al., 2019; Murè et al., 2021; Cheng et al., 2023). Another plausible explanation centers on investor preferences. A growing number of investors are integrating ESG factors into their investment decisions (Capelle-Blancard and Petit, 2019; Chen and Yang, 2020; Serafeim and Yoon, 2022, 2023), with cybersecurity increasingly recognized as a critical ESG issue. By actively pursuing ESG practices, companies with high cybersecurity risk can potentially position themselves as more attractive investment opportunities.

Concerning control variables, our findings are consistent with previous research on ESG performance (Sabbaghi, 2022; Wang et al., 2023; Chebbi, 2024; Zeng et al., 2024). Larger companies, as indicated by a positive correlation with SIZE, likely possess more resources to dedicate to sustainability initiatives, such as staffing dedicated ESG teams or implementing robust environmental management systems. Similarly, companies with higher ROA might be financially secure enough to invest in sustainable practices and technologies without compromising profitability. The positive association with LEV is less intuitive. Companies with higher debt may feel pressure from stakeholders to improve their ESG performance in order to attract environmentally and socially conscious investors or mitigate risks associated with external factors.

Table 2
Correlation matrix.

Variables	ESG	ENV	SOC	GOV	CYBER SCORE	SIZE	ROA	LEV	TANGIBLE	Cash	MARKCAP	R&D	GROWTH
ESG	1												
ENV	0.9479***	1											
SOC	0.8379***	0.7356***	1										
GOV	0.7375***	0.6115***	0.5694***	1									
CYBERISK	0.0199**	0.0141	0.0178*	0.0074	1								
SIZE	0.4580***	0.4833***	0.3555***	0.3731***	0.0041	1							
ROA	0.1292***	0.1218***	0.1222***	0.0436***	−0.0432***	−0.1473***	1						
LEV	0.1087***	0.1018***	0.1283***	0.1067***	0.0308***	0.0671***	−0.0916***	1					
TANGIBLE	−0.1073***	−0.0835***	−0.1255***	−0.0593***	−0.0108	0.0472***	−0.2241***	−0.0966***	1				
Cash	0.1160***	0.1090***	0.0749***	0.0314***	−0.0397***	−0.1371***	0.6622***	−0.2219***	−0.3734***	1			
MARKCAP	0.0869***	0.0773***	0.0858***	0.0029	−0.0286***	−0.3209***	0.6871***	−0.1410***	−0.2946***	0.6281***	1		
R&D	0.1676***	0.1916***	0.1235***	0.0600***	0.0125	−0.1214***	0.2365***	−0.1221***	−0.3608***	0.3961***	0.4343***	1	
GROWTH	−0.0842***	−0.0924***	−0.0703***	−0.0559***	0.0090	−0.0342***	0.1447***	−0.0598***	−0.0652***	0.0149	0.1528***	0.0274**	1

Note: Asterisks (*) *denote different levels of statistical significance: one asterisk* (.) represents significance at the 10 % level, two asterisks (**) indicate significance at the 5 % level, and three asterisks (***) signify significance at the 1 % level.

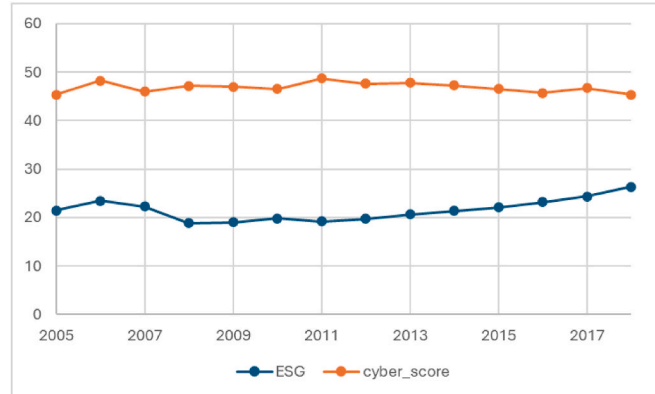

Fig. 1. Evolution of cybersecurity risk and ESG.

Table 3
The impact of cybersecurity risk on ESG performance.

VARIABLES	Fixed Effects
Intercept	6.4366** (2.43)
CYBERISK	4.1079*** (3.31)
SIZE	0.5119*** (2.86)
ROA	3.2530*** (3.08)
LEV	0.0256*** (4.48)
TANGIBLE	0.1467 (0.20)
Cash	2.6196*** (2.60)
MARKCAP	−0.1476* (−1.83)
R&D	−2.9487 (−0.80)
GROWTH	−0.0032** (−2.11)
Year_FE	YES
Industry_FE	YES
State_FE	YES
Sample Size	14,036
F_statistic	82.77***
Adjusted R^2	0.0171

Note: Asterisks (*) *indicate statistical significance at the 10 % level, double asterisks (**) denote significance at the 5 % level, and triple asterisks* (***) signify a 1 % level of significance.

Conversely, ample Cash reserves allow companies to invest in ESG initiatives without relying on external financing, potentially funding cleaner production technologies, employee training programs, or community development projects. Interestingly, the relationship between company size and ESG performance might not be linear. While MARKCAP exhibits a negative coefficient, suggesting a slight negative impact on ESG, this could indicate a curvilinear trend. Despite having resources, large companies may face challenges in implementing ESG initiatives across their complex operations. Additionally, some large companies may operate in inherently environmentally or socially impactful industries, which can affect their overall ESG score. Finally, our analysis did not reveal a statistically significant association between a company's research and development (R&D) investment or growth rate (GROWTH) and its overall ESG performance.

The F-statistic of 382.77 is highly significant (***), suggesting that the overall model is statistically significant.

4.2.2. *The moderator role of CEO duality*

Table 4 examines the influence of CEO duality on the relationship between cyber risk and ESG performance. When a CEO serves as the board chair, it can significantly alter how a company manages cybersecurity risks and addresses environmental, social, and governance (ESG) priorities. The analysis reveals an interaction effect, suggesting that the relationship between cybersecurity risk and ESG performance depends on whether the roles of CEO and board chair are combined. Specifically, companies with CEO duality tend to approach

Table 4
The role of CEO duality.

VARIABLES	Fixed Effects
Intercept	8.2481*** (4.86)
CYBERISK	3.2393** (2.58)
CEODUALITY	1.5943*** (2.82)
CYBERRISK*CEODUALITY	5.5487** (2.49)
SIZE	0.6576*** (5.74)
ROA	3.6426*** (3.67)
LEV	0.04795*** (4.11)
TANGIBLE	1.7287 (0.21)
Cash	2.5660** (2.28)
MARKCAP	0.9124*** (8.24)
R&D	−0.5965** (−2.49)
GROWTH	−0.0206*** (−3.98)
Year_FE	YES
Industry_FE	YES
State_FE	YES
Sample Size	14,036
F_statistic	78.59***
Adjusted R^2	0.0372

Note: Asterisks (*) *indicate statistical significance at the 10 % level, double asterisks (**) denote significance at the 5 % level, and triple asterisks (***)* signify a 1 % level of significance.

Table 5
The impact of cybersecurity risk on ESG pillars.

VARIABLES	ENV	SOC	GOV
Intercept	61.9497*** (5.47)	33.5494*** (4.63)	39.3935*** (11.40)
CYBERISK	6.9206*** (2.63)	4.3911*** (2.23)	1.7533* (1.84)
SIZE	2.5616*** (5.66)	1.4621*** (5.08)	0.3622*** (2.65)
ROA	4.0165* (1.68)	3.8571*** (2.30)	1.7817*** (2.20)
LEV	0.0346** (2.57)	0.0089 (0.99)	0.0092*** (2.10)
TANGIBLE	3.2146* (1.77)	−0.2938 (−0.24)	0.2228 (0.39)
Cash	5.0972** (2.03)	2.7492 (1.64)	0.7468 (0.97)
MARKCAP	−0.7737*** (−3.40)	−0.0443 (−0.34)	−0.0391 (−0.63)
R&D	3.2135 (0.26S)	−2.7150 (−0.44)	4.6822* (1.65)
GROWTH	−0.0094*** (−2.04)	−0.0047* (−4.63)	−0.0002 (−0.24)
Year_FE	Yes	Yes	Yes
Industry_FE	Yes	Yes	Yes
Sample Size	7623	11,341	14,036
F_statistic	68.21***	92.76***	51.72***
Adjusted R^2	0.0027	0.0094	0.0601

Note: Asterisks (*) *indicate statistical significance at the 10 % level, double asterisks (**) denote significance at the 5 % level, and triple asterisks (***)* signify a 1 % level of significance.

cybersecurity and ESG matters differently than those with distinct roles. This interaction effect underscores the pivotal role of CEO duality in moderating the relationship between cybersecurity risk and ESG outcomes, as its presence intensifies the positive association between the two.

Empirical evidence highlights that board characteristics, particularly CEO duality, play a moderating role by amplifying the positive effects of cybersecurity risk on ESG outcomes. CEO duality, where an individual simultaneously holds the roles of CEO and chairperson, simplifies decision-making processes, thus enabling firms to respond promptly to cybersecurity challenges (Chen et al., 2024). Streamlined governance under CEO duality facilitates efficient allocation of resources toward cybersecurity measures, enhancing organizational resilience and supporting ESG frameworks (Meeprom et al., 2024). Although CEO duality can lead to improved operational responses, concerns remain regarding its potential to weaken board independence and long-term governance quality. These concerns are particularly relevant in scenarios lacking complementary factors such as board diversity and independence (Abdullah et al., 2024). Nevertheless, the positive influence of CEO duality in cybersecurity-focused contexts demonstrates how consolidated leadership can align immediate risk management priorities with broader ESG strategies (Mubeen et al., 2021; Shackelford et al., 2023).

4.2.3. Cybersecurity risk and ESG pillars

Table 5 illuminates a fascinating dimension of our analysis: the relationship between cybersecurity risk and the individual pillars of ESG (environmental, social, and governance). The results reveal a statistically significant positive association between cyber risk (CYBERRISK) and all three ESG components: environmental performance (ENV), social performance (SOC), and governance practices (GOV).

Companies that are acutely aware of cyber threats might prioritize environmentally sustainable practices to reduce their overall risk profile. This could involve adopting energy-efficient technologies, minimizing resource consumption in IT operations, or investing in renewable energy sources (Barykin et al., 2022; Mneimneh et al., 2023). By lowering their environmental footprint, they potentially decrease their vulnerability to cyberattacks targeting critical infrastructure.

A heightened focus on cybersecurity might lead to a stronger emphasis on social responsibility practices. This could encompass investing in employee training programs on cybersecurity awareness, fostering a culture of ethical behavior, or prioritizing data privacy and security. By prioritizing the well-being of their employees and customers, companies can potentially build trust and resilience against cyberattacks that exploit social vulnerabilities. In addition, Bamiatzi et al. (2023) investigate the potential for Corporate Social Responsibility to act as a buffer for companies experiencing data breaches. Analyzing 230 breached firms reveals a crucial finding: data breaches inflict significantly harsher financial consequences on companies with low corporate social responsibility (CSR) efforts. This negative impact is particularly pronounced in industries that handle sensitive consumer data. The authors further suggest a reactive response from companies – a post-breach surge in CSR activities. This highlights a potential strategy for rebuilding lost trust and regaining stakeholder confidence.

Strong corporate governance (GOV), a key ESG component, emerges as a potential driver of the positive ESG-cybersecurity risk link. Robust GOV practices foster a risk management culture, leading to proactive identification and mitigation of cybersecurity vulnerabilities. This aligns with research by Macnish et al. (2020), Formosa et al. (2021), and Kumar et al. (2021), highlighting the crucial role of ethical governance in promoting cybersecurity awareness. Companies facing high cyber risks may prioritize strong GOV as a defense mechanism, bolstering their overall cybersecurity posture.

4.2.4. The impact of cybersecurity risk on ESG performance in each sector

Table 6 delves deeper into our analysis, revealing a fascinating interplay between cybersecurity risk and ESG performance across different industry sectors. The findings paint a compelling picture, with CYBERISK generally exerting a positive and statistically significant influence on a company's ESG performance for all sectors. This suggests that companies facing higher cybersecurity risk levels, as measured by our CYBERISK variable, tend to invest more in environmental, social, and governance practices.

However, Table 6 also highlights a crucial exception: the financial sector. Here, the relationship flips, with CYBERISK exhibiting a negative association with ESG performance. Some factors could contribute to this intriguing divergence. Financial institutions operate in a hyper-sensitive environment where trust is paramount (Uddin et al., 2020a,b; Alzoubi et al., 2022; Pollmeier et al., 2023). A data breach or cyberattack can inflict severe reputational damage, eroding consumer confidence and potentially leading to regulatory fines. This heightened risk could lead some financial institutions to prioritize short-term measures over long-term investments in ESG practices, potentially hindering their overall ESG performance. Furthermore, Financial institutions already face a complex web of regulations. The additional burden of cybersecurity compliance might limit their bandwidth and resources dedicated to broader ESG initiatives. This could create a trade-off between cybersecurity investments and investments in environmental or social

Table 6
The impact of cybersecurity risk on ESG performance in each sector.

	Communication Services	Consumer discretionary	Consumer staples	Energy	Financials	Healthcare	Industrials	Materials	Information Technology	Real State	Utilities
Intercept	−70.9535***	−92.7743***	−11.6909	−27.5335	−84.9998***	−51.0434***	−128.615***	−192.1527***	−73.7609***	−102.4961***	102.7038**
	(−10.94)	(−16.71)	(−0.66)	(−1.42)	(−22.13)	(−4.54)	(−19.27)	(−15.13)	(−6.12)	(−10.81)	(2.06)
CYBERISK	13.7643** (2.16)	5.9935* (1.67)	9.3961*	6.7970*	−2.2091**	−2.2538	0.5111** (2.15)	9.9039* (1.97)	14.1280** (2.68)	4.7867 (0.84)	8.1301*
			(1.88)	(1.98)	(−2.66)	(−0.60)					(2.02)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year_FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm_FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State_FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample Size	582	2281	811	897	2226	1683	2509	1015	467	915	678
Adjusted R^2	0.4952	0.4442	0.2060	0.1845	0.4677	0.3090	0.5007	0.5420	0.3042	0.4905	0.3434

Note: Asterisks (*) *indicate statistical significance at the 10 % level, double asterisks (**) denote significance at the 5 % level, and triple asterisks (***)* signify a 1 % level of significance.

responsibility programs. Finally, Financial institutions might prioritize investments in cyber defense mechanisms over broader sustainability initiatives (Ghelani et al., 2022). This could be driven by a short-term focus on mitigating immediate cyber threats, potentially at the expense of neglecting the long-term benefits of a holistic ESG approach.

4.2.5. *Check of robustness*

To solidify the reliability of our findings, we conducted a series of robustness checks on our initial model. This involved two main approaches. We re-estimated the model using a different measure for cybersecurity risk to ensure the results were not dependent on the specific measure chosen. We also employed various techniques to address potential endogeneity concerns, where the independent variable (cybersecurity risk) might be influenced by the dependent variable (ESG performance) and vice versa. These techniques included the Heckman two-stage and instrumental variable two-stage least squares (IV-2SLS) models.

i. Alternative measure of cybersecurity risk

To bolster the robustness of our findings, we re-ran our regression analysis using an alternative cybersecurity risk score developed in a recent study by Florackis et al. (2023). This alternative measure provides a distinct perspective on a company's vulnerability to cyber threats. The results, presented in Table 7, are reassuringly consistent with our initial findings. We observe a positive association between the new measure of cybersecurity risk and a company's ESG performance, as well as a positive moderating role of CEO duality on the cybersecurity risk-ESG nexus.

ii Endogenity

Our initial analysis revealed a positive association between cybersecurity risk and firm ESG performance. However, this association might be influenced by endogeneity issues. Endogeneity arises when the independent variable (cybersecurity risk) and the dependent variable (ESG performance) can potentially influence each other. This can occur due to reverse causality (e.g., strong ESG practices may make a company a less attractive target for cyberattacks), unobserved time-varying factors that affect both variables, or selection bias (e.g., only certain companies are included in the study). To address these concerns and ensure the robustness of our findings, we employed two established methods:

Table 7
An alternative measure of cybersecurity risk.

VARIABLES	Fixed Effects
Intercept	7.3033*** (8.19)
CYBERISK	0.7207*** (3.41)
CEODUALITY	0.5153*** (2.96)
CYBERRISK*CEODUALITY	0.675** (2.33)
SIZE	4.8158*** (85.12)
ROA	9.2162*** (5.99)
LEV	0.0182*** (4.12)
Cash	5.8660*** (5.00)
TANGIBLE	4.4326*** (4.30)
MARKCAP	1.3009*** (13.72)
R&D	26.7974*** (−5.35)
GROWTH	−0.0133*** (10.42)
Year_FE	YES
Industry_FE	YES
State_FE	YES
Sample Size	11,654
F_statistic	78.87***
Adjusted R^2	0.3651

Note: Asterisks (*) *indicate statistical significance at the 10 % level, double asterisks (**) denote significance at the 5 % level, and triple asterisks (***)* signify a 1 % level of significance.

the two-stage least squares (2SLS) model and the Heckman two-stage model.

Not all companies face the same level of cybersecurity risk. Some companies might be more likely to invest in cybersecurity due to their inherent characteristics. We follow a two-step procedure developed by Heckman (1976, 1979) to account for this potential bias. In the first step, we predict the likelihood that a firm will face a significant cybersecurity risk. We achieve this by estimating a model where a score of 1 indicates that a firm's cybersecurity risk is above the average for our sample, and a score of 0 indicates it is below average. Building upon the first stage, we re-run our primary analysis (Equation (1)) with an additional factor. This factor, known as the inverse Mills ratio (IMR), helps us account for any potential bias arising from firms' self-selection into different cybersecurity risk levels. The first stage of our analysis mirrors the variables used in the primary analysis, but with an additional twist. We also include one or more exclusion restriction variables. These additional variables are crucial because they significantly impact the likelihood of a firm having a high cybersecurity risk, but do not directly influence its ESG performance. Table 8 showcases the results of this two-step process. Notably, our findings remain unchanged even after careful consideration.

Possible bias arising from self-selection. This bolsters the confidence and The soundness of our findings concerning the positive link between cybersecurity risk and ESG performance.

We employ instrumental variables with two-stage least squares (IV-2SLS) to mitigate potential biases in our data. This methodology helps us address two main concerns: unobservable time-varying factors and reverse Causality. This approach tackles a potential issue in our data analysis: endogeneity. An ideal IV is like a bridge. It has a strong connection to cybersecurity risk but no direct impact on a company's ESG performance (the error term). Additionally, it should not be influenced by any unseen factors that might affect innovation (Wooldridge, 2019).

As an instrumental variable (IV), we use the average cybersecurity risk among firms in the same industry, not including the specific focal firm. The rationale behind this choice is that companies in the same industry often face similar cyber threats (Hossain et al., 2023; Hsu et al., 2015).

To ensure the reliability of our chosen instrument, we conducted a Wald chi-squared test. This test resulted in a value of 70.44, which is significantly higher than the critical value. This strong result indicates that PEER is a valid instrument and effectively addresses concerns about endogeneity. Moving on to the second stage of the IV-2SLS analysis, presented in Table 7, we observe a positive relationship between cybersecurity risk and ESG performance. This finding holds even after accounting for potential endogeneity issues, further strengthening the validity of our initial conclusion.

Table 8
Two-step regression of 2SLS and Heckman methods.

VARIABLES	2SLS	Heckman
Intercept	−51.9195*** (−8.24)	−107.6657*** (−17.14)
CYBERISK	7.9955*** (6.19)	4.4989*** (3.68)
SIZE	4.4141*** (28.50)	4.6290*** (31.58)
ROA	10.5583** (2.53)	131.1084*** (3.20)
LEV	0.0586*** (4.85)	0.0371*** (3.18)
TANGIBLE	5.7440*** (5.88)	7.0984*** (2.87)
Cash	−2.5884 (−0.67)	0.2456 (0.06)
MARKCAP	0.8568*** (3.47)	0.9942*** (4.14)
GROWTH	−0.0236*** (−3.30)	−0.0280*** (−3.49)
R&D	60.7960*** (8.21)	42.1318*** (6.60)
Year_FE	YES	YES
Industry_FE	YES	YES
State_FE	YES	YES
Sample Size	15,456	15,534
Wald chi2	70.44***	98.20***

Note: Asterisks (*) *indicate statistical significance at the 10 % level, double asterisks (**) denote significance at the 5 % level, and triple asterisks (***)* signify a 1 % level of significance.

5. Conclusion

This study examines the relationship between cybersecurity risk and ESG performance, with a focus on the moderating role of CEO duality within corporate governance. Drawing on a large panel of U.S.-listed firms, the results consistently show that cybersecurity risk acts as a strategic driver of ESG enhancement, particularly when firms face external pressures to demonstrate resilience, transparency, and accountability. The findings position cybersecurity not only as a technological concern but also as a governance and sustainability issue that influences how organizations pursue and communicate their ESG commitments.

A key contribution of this study lies in uncovering the sector-specific dynamics of this relationship. While most industries exhibit a positive correlation between cybersecurity risk and ESG performance, the financial sector stands out as an exception, where this association becomes negative. This divergence suggests that firms in finance, operating under intense regulatory oversight and public trust, may prioritize short-term compliance or defensive strategies over broader ESG investments when facing cyber threats. In high-stakes environments, the need for immediate risk containment may overshadow ESG efforts, or firms may struggle to balance cybersecurity demands with their broader sustainability goals. This nuanced insight underscores the importance of tailoring ESG strategies to industry-specific risk profiles and operational constraints. It also calls for more robust integration of cybersecurity governance into ESG frameworks, particularly in sectors with the highest reputational and regulatory stakes.

Furthermore, the study highlights the role of CEO duality as a governance lever that can magnify the positive effects of cybersecurity awareness on ESG performance. When CEOs also serve as board chairs, firms appear better positioned to swiftly align cybersecurity measures with sustainability goals, especially in industries where rapid strategic responses are critical. However, this governance arrangement must be accompanied by strong accountability mechanisms to avoid potential risks associated with power concentration.

Overall, this research contributes to the literature by conceptualizing cybersecurity as a governance-linked ESG determinant and identifying CEO duality as a context-dependent moderator. For practitioners, the findings offer important implications: Firms should view cybersecurity governance not in isolation, but as part of an integrated ESG strategy—one that adapts to sector-specific challenges. In sectors such as finance, this may require rethinking traditional ESG approaches to ensure they are resilient, adaptive, and capable of managing digital-era vulnerabilities without compromising long-term sustainability commitments.

Future research could further explore how industry characteristics shape the cybersecurity-ESG nexus and examine whether evolving regulatory frameworks and disclosure standards help mitigate the trade-offs observed in sensitive sectors such as finance.

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Declaration of competing interest

I declare i have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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